

Def. Let $A \in M_{m \times n}(F)$ be given where $F = \mathbb{R}$ or $\mathbb{C}$. Define the supremum norm of A

$$\|A\|_p \stackrel{\text{def}}{=} \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} \stackrel{(*)}{=} \sup_{\|x\|_p=1} \|Ax\|_p$$

where $\|\cdot\|_p$ is the p -norm, that is,

$$\|\cdot\|_p: F^n \rightarrow \mathbb{R}; (a_i)_{i=1}^n \mapsto \left(\sum_{i=1}^n |a_i|^p \right)^{\frac{1}{p}}.$$

The equality $(*)$ is given by the fact that for any non-zero vector x , then

$$\frac{\|Ax\|_p}{\|x\|_p} = \left\| A \cdot \frac{x}{\|x\|_p} \right\|_p = \|Ax'\|_p, \text{ where } \|x'\|_p = 1.$$

In fact, the value $\|A\|_p$ is the smallest value C such that

$$\|Ax\|_p \leq C \cdot \|x\|_p \text{ for all non-zero } x \in F^n$$

That is, given $A \in M_{m \times n}(F)$ and $x \in F^n$,

$$\|Ax\|_p \leq \|A\|_p \|x\|_p$$

Prop. Given $A \in M_{m \times n}(F)$ where $F = \mathbb{R}$ or $\mathbb{C}$,

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |A_{ij}| \quad \text{and} \quad \|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}|$$

Proof. Given $x \in \mathbb{R}^n$ with $\|x\|_1 = 1$, denote $x = (x_1, \dots, x_n)$.

$$\|Ax\|_1 = \underbrace{\left\| \sum_{j=1}^n x_j \cdot A e_j \right\|_1}_{\text{def}} \leq \underbrace{\sum_{j=1}^n |x_j| \cdot \|A e_j\|_1}_{\text{Hom + Sub. add}} \leq \sum_{j=1}^n |x_j| \cdot \max_{1 \leq i \leq n} \|A e_j\|_1 = \max_{1 \leq j \leq n} \|A e_j\|_1.$$

And, take $x = e_k$ such that e_k maximizes the $\|A e_j\|_1$, then we obtain the first.

Given $x \in \mathbb{R}^n$ with $\|x\|_\infty = 1$. Then, for any $1 \leq k \leq m$,

$$|(Ax)_k| = \left| \sum_{j=1}^n A_{kj} \cdot x_j \right| \leq \sum_{j=1}^n |x_j| \cdot |A_{kj}| \leq \sum_{j=1}^n |A_{kj}|$$

and the equality holds when $x = (1, 1, \dots, 1)$. So proved. $\square$

The $\|\cdot\|_2$ is so special, because it is induced by the inner product.

Note that the fact that for any unitary $Q \in M_{m \times n}(F)$, Q preserves the norm. That is,

$$\|Qx\|_2^2 = \langle Qx, Qx \rangle = \langle x, x \rangle = \|x\|_2^2$$

Thm. Given $A \in M_{m \times n}(F)$ where $F = \mathbb{R}$ or $\mathbb{C}$,

$$\|A\|_2 = \sigma_{\max}, \text{ the largest singular value of } A.$$

Proof. Given A , there is a Singular Value decomposition,

$$A = U \Sigma V^* \text{ where } U \in M_{m \times m}, V \in M_{n \times n} \text{ are unitary,}$$

$$\text{and } \Sigma = \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{pmatrix} \in M_{m \times n}, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$$

so,

$$\|Ax\|_2 = \|U \Sigma V^* x\|_2 = \|\Sigma V^* x\|_2 = \|\Sigma y\|_2 \text{ where } y = V^* x$$

Since the unitary operation is isomorphism, and isometry, we can replace $V^* x$ into y .

And, the p-norm of diagonal matrix is the largest absolute value of its entry, so proved.

Problems.

For $A = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$, Compute $\|A\|_1, \|A\|_\infty, \|A\|_F, \|A\|_2$.

Proof.

$$\|A\|_1 = |-2| + |1| = 3, \quad \|A\|_\infty = |1| + |-2| = 3,$$

$$AA^* = \begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$$

$$\|A\|_F = \sqrt{1+4+1+4} = \sqrt{10} \quad (= \sqrt{\text{tr}(AA^*)}).$$

Meanwhile, $A^*A = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 2 \\ 0 & 2 & 4 \end{bmatrix}$ has a characteristic polynomial

$$\chi_{A^*A} = t(t-3)(t-7)$$

$$\text{So, } \|A\|_2 = \sqrt{7}.$$

Prove that Frobenius Cauchy-Schwarz inequality

$$|\text{tr}(A^T B)| \leq \|A\|_F \cdot \|B\|_F.$$

And solve $\max_{\|X\|_F \leq 1} \text{tr}(A^T X)$, find all maximizers for a non-zero A .

Rayleigh quotient.

Def. Let B be a n by n Hermitian matrix.

The Rayleigh quotient for $X \neq 0$ is

$$R(X) \stackrel{\text{def}}{=} \frac{\langle BX, X \rangle}{\|X\|^2}$$

Thm. Let B be a n by n Hermitian matrix.

Then, the spectrum $\sigma(B)$ is non-empty real numbers set and

$$\max_{X \neq 0} R(X) = \max \sigma(B) \quad \text{and} \quad \min_{X \neq 0} R(X) = \min \sigma(B).$$

Proof. Let $\beta = \{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis for B such that each α_i is eigenvector of B corresponding to $\lambda_i \in \sigma(B)$.

Now, given $X \in F^n \setminus \{0\}$, $X = \sum_{i=1}^n a_i \cdot \alpha_i$ for some $a_i \in F$, so that

$$\begin{aligned} R(X) &= \frac{\langle BX, X \rangle}{\|X\|^2} = \frac{\langle \sum a_i \cdot \lambda_i \cdot \alpha_i, \sum a_i \cdot \alpha_i \rangle}{\|X\|^2} \\ &= \left(\sum_{i=1}^n \lambda_i \cdot |a_i|^2 \right) / \left(\sum_{i=1}^n |a_i|^2 \right) \\ &\leq \max \sigma(B) \cdot \left(\sum_{i=1}^n |a_i|^2 \right) / \left(\sum_{i=1}^n |a_i|^2 \right), \end{aligned}$$

So, the inequality is obtained, and take $X = \alpha$ such that $\alpha \in \beta$ satisfying $B\alpha = \max \sigma(B)\alpha$, so we obtain $R(\alpha) = \max \sigma(B)$.
Similarly, the minimum can be obtained. $\square$

Lemma. Let A be an any square matrix over $\mathbb{C}$ or $\mathbb{R}$.

Then, λ is eigenvalue of $A^*A \Leftrightarrow \lambda$ is eigenvalue of AA^* .

Proof. Let $\lambda \in \mathbb{F}$ be an eigenvalue of A^*A .

If $\lambda = 0$, then $\det(A^*A) = \det A^* \cdot \det A = \det(AA^*) = 0$, so

AA^* has 0 as an eigenvalue. And the Converse is similarly shown.

If $\lambda \neq 0$, then for some non-zero vector X , $A^*AX = \lambda X$. Then,

$$AA^*AX = A(\lambda X) = \lambda AX. \text{ But, if } AX = 0, \text{ then } A^*AX = 0 = \lambda X, \\ \text{so } AX \neq 0.$$

Therefore, AA^* has λ as an eigenvalue. The Converse can be shown, exactly same way.

Corollary. Let $A \in M_{nn}(\mathbb{F})$ be an invertible matrix. Then,

$$\|A^{-1}\| = \frac{1}{\sigma_{\min}(A)}$$

Proof. Let $\lambda_1 \geq \dots \geq \lambda_n$ be eigenvalues of A^*A . By the Lemma above, these are also eigenvalues of AA^* . And, since

$$\|A^{-1}\|^2 = \text{the largest eigenvalue of } (A^{-1})^*(A^{-1}) = (AA^*)^{-1},$$

and $(AA^*)^{-1}$ has eigenvalues as

$$\lambda_1^{-1} \geq \dots \geq \lambda_n^{-1}, \quad (\text{since } BX = \lambda X \Rightarrow \lambda^{-1} = B^{-1}\lambda, \text{ if } \det B \neq 0)$$

$$\text{so } \|A^{-1}\| = \sqrt{\lambda_n^{-1}} = 1/\sigma_{\min}(A). \quad \square$$